

Block-diagonal block-encoding: Clader-Frobenius vs SVD interferometer

Task: block-encode the block-diagonal operator $A_{\text{block}} = \sum_k |k\rangle\langle k| (x) A_k$ with $K = 8$ blocks, each A_k of size $N \times N$, $N = 2^n$, $n = 4..10$, $\text{phase_bitsize} = 32$. Two methods, each Toffoli(T)-optimal and qubit-optimal.

- (1) CLADER-FROBENIUS BLOCK (arXiv:2206.03505 generalization)
Encodes $\sum_k |k\rangle\langle k| (x) A_k / F_{\text{max}}$, $F_{\text{max}} = \max_k ||A_k||_F$. A read-only k register addresses every QROAM:
- flag : k -controlled rotation $|0\rangle_f \rightarrow (F_k/F_{\text{max}})|0\rangle + \sqrt{1-F_k^2/F_{\text{max}}^2}|1\rangle$
 - U_L : k -controlled prep of $|\phi_k\rangle = \sum_j ||(A_k)_{\{j,.\}}||/F_k |j\rangle$ (+ swap)
 - U_R : (k,j) -controlled prep of $|\psi_{\{k,j\}}\rangle = \sum_l (A_k)_{\{j,l\}}/|| (A_k)_{\{j,.\}} || |l\rangle$
- Post-selecting $f=0$ and the prep ancilla on $|0\rangle$ yields $(A_k)_{\{j,l\}}/F_{\text{max}}$.
T-opt : Toffoli-minimal QROAM select-swap batch (more ancilla).
Q-opt : un-batched $\lambda = 1$ QROAM (fewest qubits, more T).
- (2) SVD INTERFEROMETER BLOCK ($\alpha = 1$)
 $A_k = U_k \Sigma_k V_k$; U_k, V_k via block-unitary interferometer, Σ_k via QROAMClean + ctrl-AddIntoPhaseGrad + QROAMCleanAdjoint. Each metric optimized over a QROAM batching λ (blocking the K -block / pair dimension).

SCALING FITS $y = c \cdot 2^{(a \cdot n)}$ (slope a per doubling of N ; K fixed):

Clader-Frobenius block, Toffoli-opt	T: $a=1.00$	Q: $a=0.96$
Clader-Frobenius block, qubit-opt	T: $a=1.98$	Q: $a=0.06$
SVD interferometer block, Toffoli-opt	T: $a=1.40$	Q: $a=0.40$
SVD interferometer block, qubit-opt	T: $a=1.73$	Q: $a=0.03$

KEY CAVEAT - subnormalization is NOT equal:

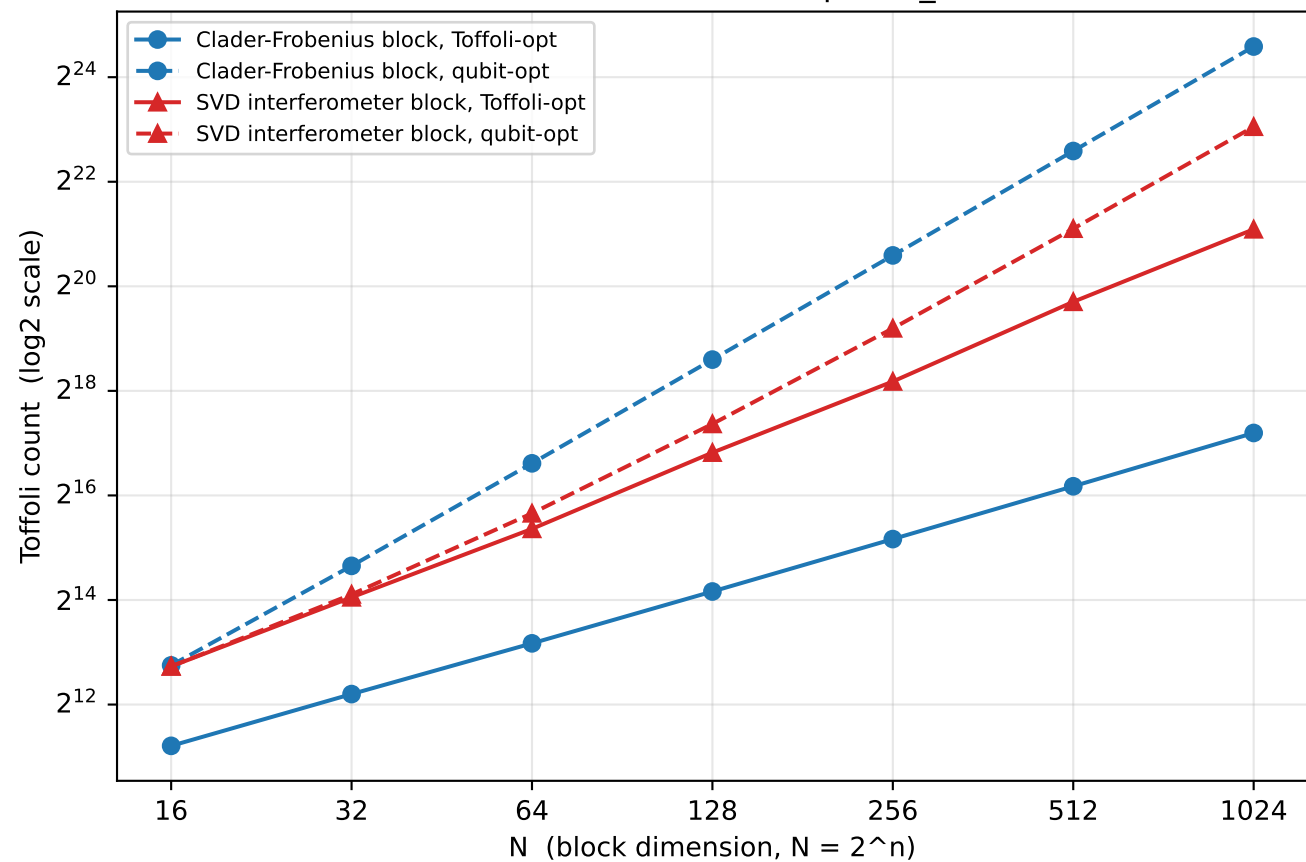
- * Clader-block: $\alpha = F_{\text{max}} = \max_k ||A_k||_F$ ($\sim N$ for dense $O(1)$ blocks, independent of K). SVD-block: spectral-norm-optimal $\alpha = 1$.
- * Downstream algorithms (QSVT, QPE, qubitization) need $\sim \alpha$ block-encoding queries; the EFFECTIVE Clader cost is per-query Toffoli times F_{max} ($\sim N$).
- * Read the raw T/Q as the cost of ONE block-encoding query. The SVD method pays more per query but needs far fewer queries; Clader is cheap per query but its large α must be amortized. Choice is application-dependent.

Both: error $\sim O(2^{-\text{phase_bitsize}})$; counts from Qualtran QECGatesCost / QubitCount.

Toffoli (T) / qubit (Q) counts: block-diagonal $\sum_k |k\rangle\langle k|$ (x) A_k , $K = 8$
 phase_bitsize = 32; $\alpha(\text{Clader}) = F_{\text{max}}$, $\alpha(\text{SVD}) = 1$

n	N	Clader-Frobenius Toffoli-opt	Clader-Frobenius qubit-opt	SVD interferometer Toffoli-opt	SVD interferometer qubit-opt
4	16	T=2,369 Q=309	T=6,886 Q=88	T=6,793 Q=109	T=6,793 Q=109
5	32	T=4,706 Q=568	T=25,758 Q=92	T=16,972 Q=174	T=17,641 Q=111
6	64	T=9,221 Q=1,083	T=100,310 Q=96	T=42,060 Q=176	T=51,689 Q=113
7	128	T=18,336 Q=2,110	T=396,814 Q=100	T=115,602 Q=305	T=169,065 Q=115
8	256	T=36,739 Q=4,161	T=1,579,590 Q=104	T=296,850 Q=307	T=600,681 Q=117
9	512	T=73,986 Q=8,260	T=6,304,382 Q=108	T=853,790 Q=564	T=2,250,857 Q=119
10	1024	T=150,025 Q=16,455	T=25,191,094 Q=112	T=2,233,118 Q=566	T=8,697,961 Q=121

Toffoli count vs N (K = 8 blocks, phase_bitsize = 32)



Qubit count vs N (K = 8 blocks, phase_bitsize = 32)

